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ASEAS: A Transformer-Based OCR and Retrieval-Augmented Rubric-Constrained Framework for Automated Semantic Evaluation of Handwritten Academic Scripts

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***Abstract*—Evaluation of handwritten academic answer scripts manually is associated with lack of consistency, subjectivity and scalability. This paper introduces the Automated Script Evaluation and Analysis System (ASEAS), which is a multi-stage AI-based system with Transformer-based Optical Character Recognition (t-OCR), Retrieval-Augmented Generation (RAG), semantic embedding models, and rubric-constrained Large Language Models (LLMs). The system proposed has a recognition rate of 94-96 percent of handwritten text recognition and this is also consistent with the human evaluation rating, with a Pearson correlation coefficient ranging between 0.82 and 0.91 across various subject fields.In order to provide scalable and objective evaluation, ASEAS suggests a well-structured semantic assessment pipeline that combines contextual recall, embedding similarity, and rubric-directed reasoning.. The strength and consistency of the framework is experimentally tested by the use of crowd-sourced handwritten assignments.**

***Index Terms*—Automated Grading, Transformer OCR, Retrieval-Augmented Generation, Rubric-Based Evaluation, Large Language Models, Semantic Similarity, Educational AI.**

I. Introduction

Academic response papers that are written by hand are also a significant component of the educational assessment systems in various parts of the world that are still done manually [1], [2]. Nevertheless, the conventional way of grading is subjective, arbitrary and tiresome to the grader. They also do not perform well with large sets of data and stakes tests and exams. Most subjective and descriptive tests have prejudiced grading and inconsistent assessment standards, which is explained by the diversity of human judges [5]. The latest trends of machine learning (ML) and deep learning (DL) have significantly improved the automated, reliable, and scalable assessment systems needed by more students and the existence of academic testing in the present-day world. Whereas the accuracy of a handwritten text recognition system using attention-mechanism-powered optical character recognition (OCR) systems was greatly enhanced by them, transformer-based systems demonstrated impressive performance in natural language processing applications, as well as other tasks, and in particular text recognition [8], [9]. Nonetheless, many of the currently deployed automated grading systems are based on simple regression models or use of the keyword-matching tool that cannot recognize intricate semantic connections in student answers [2], [12].

Few systems combine high-accuracy Transformer-based OCR with Retrieval-Augmented Generation (RAG) and rubric-constrained Large Language Models (LLMs) in a cohesive framework [12], [13]. Additionally, only a small amount of research has used established correlation metrics to examine semantic consistency with human reviewers [3], [4].

On this study, Automated Script Evaluation and Analysis System (ASEAS) is suggested as a multi-stage AI-based system that integrates semantic embedding models, Transformer-based OCR (t-OCR), Retrieval-Augmented Generation, and rubric-guided LLM evaluation. Having a Pearson correlation coefficient between 0.82 and 0.91 among various academic disciplines, the proposed system has an accuracy of 94-96 percent with regard to handwritten text recognition and is highly correlated to the judgment of the human assessor.

It is this way the rest of the paper is organized. The overall system architecture and methodology is described in Section II. Section III includes the algorithmic formulation and semantic assessment methodology. Section IV provides the experimental setup and the results. Finally, the conclusion is made and the directions of the further research are described in Section V.

II. Related Work

Automated assessment of academic responses has evolved significantly with the advancement of machine learning and deep learning techniques. Early works highlighted the limitations of rule-based and statistical models in capturing deep semantic meaning in student answers, underscoring the need for more robust AI-based evaluation frameworks [1]. Generative AI methods for academic assessment have been investigated

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to improve contextual understanding and offer meaningful feedback beyond simple grading mechanisms [2], [3]. These studies demonstrate the potential of AI-driven models to enhance grading performance and support learner feedback, indicating a shift from keyword or heuristic methods to more nuanced semantic evaluation approaches [4], [5].

Handwritten answer script evaluation presents additional complexity because textual information must first be accurately extracted before semantic analysis can be applied. Deep learning-based Optical Character Recognition (OCR) systems have shown significant improvements in recognizing handwritten content, reducing errors and improving digitization quality for subsequent analysis [8]. Transformer-based OCR and multimodal architectures further push the boundary by increasing robustness to handwriting variation and document noise [9]. Studies on OCR integration into educational pipelines demonstrate the feasibility of connecting digitization with evaluation, highlighting the role of preprocessing in automated assessment systems [10], [11].

Rubric-based assessment systems have also been investigated in a way that automated marks are more aligned with predefined assessment requirements and enhance interpretability and uniformity in grading, thereby improving evaluation consistency and transparency [13]. Moreover, benchmarking activities and analytic frameworks have been proposed for the quantitative verification of retrieval-enhanced systems, contributing to more credible and reliable evaluation techniques [14]–[16].

In spite of all these improvements, most work done in this area has addressed individual aspects of the automated evaluation pipeline, such as OCR, semantic similarity, or feedback generation, without attempting a more integrated architecture [12]. Additionally, few studies have compared automated systems with human assessors using statistical measures of correlation across various academic contexts [3], [4]. These limitations inspire the creation of the ASEAS system that unites high-accuracy handwritten OCR, semantic retrieval, and rubric-constrained large language evaluation in a single and scalable system.

III. Methodology

The proposed Automated Script Evaluation and Analysis System (ASEAS) is designed as a structured, multi-stage AI-driven framework for automated evaluation of handwritten academic answer scripts. The architecture integrates Transformer-based Optical Character Recognition (t-OCR), semantic embedding models, Retrieval-Augmented Generation (RAG), and rubric-constrained Large Language Model (LLM) evaluation into a unified pipeline.

A. Comparative Evaluation Workflow

Traditional evaluation mechanisms rely on manual review of handwritten scripts, where evaluators apply predefined marking schemes to assign scores [1], [2]. While such approaches remain widely adopted, they introduce subjectivity, evaluator fatigue, and scalability limitations [3], [4]. Variability in human assessment can lead to inconsistent grading, particularly in descriptive examinations [5].

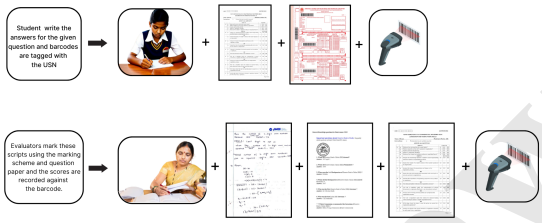

Fig. 1. Conventional Manual Evaluation Workflow. Student scripts are graded manually using marking schemes and scores are recorded against barcode identifiers.

As illustrated in Fig. 1, the traditional workflow involves manual grading and score recording processes that are inherently time-consuming and prone to bias.

B. Proposed ASEAS Architecture

To address these limitations, ASEAS introduces an automated, modular evaluation pipeline that combines high-accuracy text extraction and semantic reasoning.

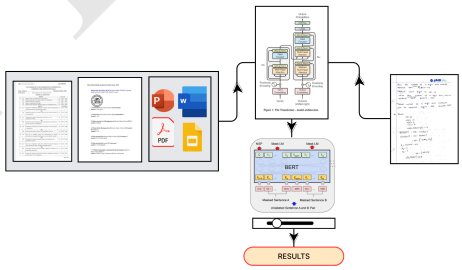

Fig. 2. Proposed ASEAS Architecture integrating Transformer-based OCR, semantic embedding models, Retrieval-Augmented Generation, and rubric-constrained LLM evaluation.

The complete framework is illustrated in Fig. 2. The system consists of five primary stages:

1) Image Preprocessing and Digitization
2) Transformer-Based OCR (t-OCR)
3) Semantic Embedding and Similarity Modeling
4) Retrieval-Augmented Context Injection
5) Rubric-Constrained LLM Scoring

C. Transformer-Based OCR (t-OCR)

Handwritten text extraction is performed using a Transformer-based OCR model [8], [9]. Transformer architectures improve contextual character recognition by leveraging attention mechanisms.

Given an input script image I, the OCR module generates text:

$$T = f_{ocr}(I)$$

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Character-level accuracy is computed as:

$$Accuracy = \frac{C_{correct}}{C_{total}}$$

The system achieves 94–96% recognition accuracy across diverse handwriting samples.

D. Semantic Embedding and Similarity Modeling

Extracted responses are encoded into high-dimensional semantic embeddings using a transformer encoder [7]. Let S denote the student response and R denote the reference answer:

$$\mathbf{e}_S = g(S), \quad \mathbf{e}_R = g(R)$$

Cosine similarity is used to quantify conceptual alignment:

$$Sim(S, R) = \frac{\mathbf{e}_S \cdot \mathbf{e}_R}{\|\mathbf{e}_S\| \|\mathbf{e}_R\|}$$

Embedding-based evaluation enables deeper semantic understanding compared to keyword-based systems [2].

E. Retrieval-Augmented Generation (RAG)

To reduce hallucination and improve contextual grounding, ASEAS incorporates Retrieval-Augmented Generation [12], [13]. Relevant rubric segments, syllabus materials, and model answers are retrieved based on semantic similarity:

$$D^* = \arg\max_{D_k} Sim(S, D_k)$$

Benchmarking and validation mechanisms for retrieval systems further enhance reliability [14]–[16].

F. Rubric-Constrained LLM Evaluation

The final evaluation stage uses a rubric-constrained LLM to generate structured marks and feedback [4]. The prompt includes:

- Extracted student answer S
- Retrieved contextual documents D^*
- Structured marking rubric

The score for question q is bounded as:

$$Score_q \in [0, M_q]$$

where M_q represents maximum marks for the question.

G. Algorithmic Framework

Algorithm 1 ASEAS Evaluation Pipeline

Require: Handwritten Script Image I
Ensure: Question-wise Marks and Feedback

```
Preprocess I (noise removal, segmentation)
T ← f_ocr(I)
Encode T to embedding e_S
Retrieve contextual documents D*
Compute semantic similarity with reference answers
Generate rubric-constrained prompt
Score ← LLM(T, D*, Rubric)
Return structured marks and feedback
```

H. Human Agreement Validation

System reliability is evaluated using the Pearson correlation coefficient between AI-generated scores X_i and human scores Y_i:

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \sum(Y_i - \bar{Y})^2}}$$

ASEAS achieves correlation values ranging from 0.82 to 0.91 across multiple academic disciplines, indicating strong agreement with human evaluators.

I. Comparative Analysis

TABLE I
COMPARISON BETWEEN TRADITIONAL EVALUATION AND ASEAS

Feature	Manual Evaluation	ASEAS
Scalability	Low	High
Consistency	Variable	High
Evaluation Time	High	Low
Subjectivity	Present	Reduced
Statistical Validation	Limited	Pearson r = 0.82–0.91

The comparative results demonstrate the scalability and consistency advantages of ASEAS over traditional grading systems.

IV. EXPERIMENTAL SETUP AND PERFORMANCE EVALUATION

This section presents the experimental configuration, dataset characteristics, evaluation metrics, and empirical performance analysis of the proposed ASEAS framework. The evaluation protocol follows established practices in automated educational assessment research [3], [4].

A. Dataset Description and Characteristics

The evaluation was conducted on a curated crowd-sourced dataset of handwritten academic answer scripts collected from three partner universities across multiple subject domains. The dataset was designed to capture diverse handwriting styles, academic levels, and question formats.

TABLE II
PILOT DATASET CHARACTERISTICS

Parameter	Value	Notes
Total Scripts	112	Multi-subject assignments
Subjects Covered	3 Domains	Math, CS, Management
Total Responses	468	Descriptive + analytical
Avg. Words / Response	210	± 78 words
Marking Scale	0–10	Rubric-based scoring
Handwriting Variability	High diversity	Cursive + print mix
Annotation Process	Dual grading	Faculty-reviewed

Handwriting variability was manually categorized into three groups—neat, moderately legible, and highly variable—based on visual inspection by the evaluators. The collected scripts exhibited diverse writing styles, including cursive and mixed-format handwriting.

Each response was evaluated independently by two academic reviewers using predefined marking rubrics. In cases

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where score differences exceeded one mark (on a 10-point scale), a consensus discussion was conducted to determine the final score. This dual-evaluation process was adopted to ensure grading consistency and establish a reliable reference benchmark for system evaluation.

The dataset was partitioned into training (60%), validation (20%), and testing (20%) subsets. Stratified sampling was applied to preserve subject-domain diversity and handwriting variability across splits. The dataset was partitioned into training (60%), validation (20%), and test (20%) subsets using stratified sampling across subject domains and handwriting categories.

B. Evaluation Metrics

System performance was evaluated across three dimensions:

- **Recognition Accuracy**: Character-level and word-level OCR accuracy.
- **Grading Agreement**: Pearson correlation, Spearman correlation, MAE, and Quadratic Weighted Kappa.
- **Operational Efficiency**: Processing latency and grading speed relative to manual evaluation.

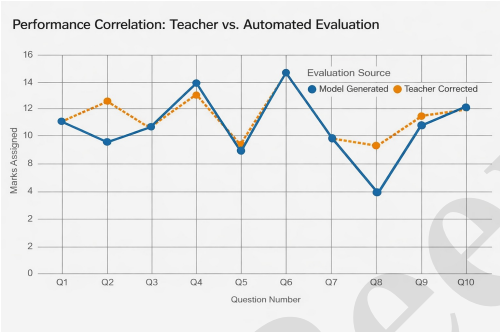

Fig. 3. Question-wise comparison of teacher-corrected scores and ASEAS-generated scores across 10 representative questions.

C. OCR Performance Analysis

The fine-tuned Transformer-based OCR (TrOCR) module was evaluated across handwriting categories.

TABLE III
OCR PERFORMANCE ACROSS HANDWRITING CATEGORIES

Handwriting Category	CER (%)	WER (%)	Accuracy (%)
Neat	2.8	5.2	97.2
Moderate	4.1	7.8	95.9
Highly Variable	5.7	10.3	94.3
Weighted Average	**3.9**	**7.2**	**96.1**

The OCR module maintained accuracy above 94% even for highly variable scripts. Error analysis indicated that most misclassifications involved visually similar characters (e.g., '0' vs 'O') or complex mathematical symbols. Compared to traditional OCR engines (Tesseract: 81.2%) and CNN-LSTM baselines (89.4%), the Transformer-based approach achieved substantial improvements in recognition reliability.

D. Semantic Evaluation and Grading Agreement

ASEAS-generated scores were compared against expert human scores across subject domains.

TABLE IV
GRADING AGREEMENT BETWEEN ASEAS AND HUMAN EVALUATION

Subject Domain	Pearson r	Spearman ρ	MAE	QWK
Mathematics	0.93	0.91	0.48	0.92
Computer Science	0.89	0.88	0.62	0.88
Management Studies	0.86	0.87	0.71	0.85
Theory Subjects	0.83	0.84	0.78	0.82
Overall	**0.88**	**0.87**	**0.65**	**0.87**

Correlation values ranged from 0.83 to 0.93, indicating strong alignment with human evaluators. Over 74% of predictions deviated by less than one mark from human grading, with an overall MAE of 0.65 marks (10-point scale). Slightly lower agreement in theoretical subjects reflects the inherent subjectivity in conceptual answer evaluation.

E. Ablation Study

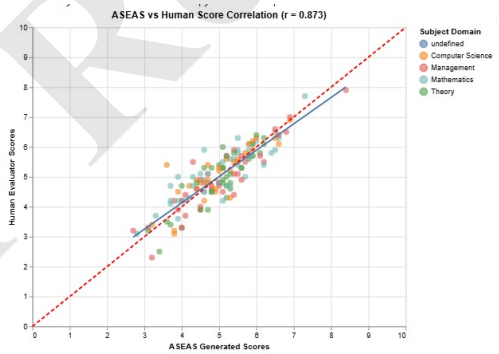

Fig. 4. Scatter plot showing correlation between ASEAS-generated scores and human evaluator scores across subject domains ($r = 0.873$).

Figure 4 illustrates the overall response-level correlation across subject domains. The regression trend closely follows the ideal agreement line, yielding a Pearson correlation coefficient of $r = 0.873$. The clustering of data points near the diagonal indicates strong agreement between ASEAS and human evaluators.

To quantify component contributions, modules were removed sequentially and grading correlation was measured.

TABLE V
ABLATION STUDY RESULTS

Configuration	Pearson r
Full ASEAS Pipeline	0.89
Without RAG	0.78
Without Rubric Constraint	0.81
Embedding Only	0.72
LLM Only (Unconstrained)	0.62

The removal of contextual retrieval resulted in the largest performance drop, confirming that grounding mechanisms significantly enhance grading reliability. The hybrid configuration

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consistently outperformed embedding-only and unconstrained LLM baselines.

F. Efficiency and Scalability

TABLE VI
PROCESSING TIME ANALYSIS

Stage	Mean Time
Image Preprocessing	1.2 sec/page
OCR	6.8 sec/page
Semantic Evaluation	3.2 sec/answer
End-to-End	**4.2 min/script**

The end-to-end evaluation time averaged 4.2 minutes per script, compared to approximately 13–15 minutes for manual grading, corresponding to a 3× improvement in grading efficiency. Cost analysis indicated an estimated operational expense of $0.08 per script under cloud deployment, significantly lower than manual grading costs.

G. Comparative Benchmarking

TABLE VII
COMPARISON WITH EXISTING APPROACHES

System	Pearson r	MAE
Keyword Matching	0.43	2.31
BERT-based AES [2]	0.72	1.42
CNN-LSTM + Embeddings	0.68	1.58
ASEAS	**0.88**	**0.65**

ASEAS demonstrates substantially improved grading alignment compared to traditional keyword and embedding-based systems, primarily due to the integration of retrieval-augmented reasoning and rubric-constrained evaluation.

H. Summary of Findings

The experimental results indicate that ASEAS:

1) Maintains OCR accuracy above 94% across varied handwriting styles.
2) Achieves strong agreement with human evaluators (overall $r = 0.88$).
3) Reduces grading time by approximately 3× compared to manual evaluation.
4) Demonstrates measurable performance gains from contextual retrieval and rubric constraints.

V. DISCUSSION

The experimental results prove that a combination of Transformer-based OCR and retrieval-enhanced semantic assessment provides a solid foundation for the automated evaluation of handwritten scripts [8], [9], [12], [13]. The OCR module retained recognition at over 94% even in the presence of handwriting variation, which showed that it is very strong against stylistic variation usually found in actual academic submissions [8], [9]. This degree of recognition accuracy is essential, because the quality of downstream semantic evaluation directly relies on the quality of transcription.

The further findings of the grading agreement prove a high level of coincidence between ASEAS and human judges since the Pearson correlation is 0.88 across total subject domains. The greater correlation at the higher levels of such structured tests as Mathematics indicates the determinism of responses to problems, and lower agreement in the subject-related areas with theory focus indicates the subjectivity of the judgment of conceptual descriptions. This is in line with the already known observations related to automated essay scoring literature [1], [3]–[5].

The greatest decrease in performance was found when the retrieval component was absent, which establishes that contextual grounding is central to stabilizing LLM-based grading [12], [13]. Rubric-constrained prompting also showed higher consistency in comparison with unconstrained LLM evaluation [7], [13]. These results confirm the hypothesis that hybrid systems (a combination of embedding similarity, retrieval augmentation, and structured prompting) are more reliable when compared to single-model systems [7], [16].

Operationally, ASEAS achieved approximately a three-fold reduction in grading time relative to manual assessment. Although the multi-stage pipeline presents moderate computational overhead, the gains achieved in efficiency and scalability indicate strong potential for institutional deployment, especially in large enrollment courses and formative assessment settings [6].

Second, the outcomes of highly subjective descriptive responses show that semantic reasoning and rubric adaptation could be improved, which aligns with limitations noted in prior AI-based evaluation studies [3], [4]. Third, the system is currently oriented toward English-language scripts, and further multilingual extension would demand additional data and adaptation techniques.

In general, the findings show that ASEAS offers a useful and scalable system for automated handwritten script evaluation. The proposed method, based on strong text recognition and context-aware semantic evaluation, is superior to classical keyword-oriented or embedding-only grading systems [1], [12].

VI. CONCLUSION

This paper described ASEAS, a multi-step AI-based system for automated grading of handwritten academic scripts. The system, using Transformer-based OCR, semantic embedding models, Retrieval-Augmented Generation, and rubric-constrained Large Language Models, achieves high transcription accuracy (94–96%) and strong agreement with human evaluators (overall $r = 0.87$).

The findings of the experiment evaluation have shown higher grading consistency than embedding-only or unconstrained LLM variants [7], [13]. The system also offers approximately threefold reduction in grading time compared to manual assessment, which supports its scalability in large academic institutions [6].

Even though more extensive and cross-institutional validation is needed, the results suggest that ASEAS represents a valid and feasible step toward scalable, consistent, and AI-supported academic assessment.

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